

Experts Problem and Online Convex Optimization

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References

- Lecture 1 and 2 of [Haipeng's course](#)

Learning from Experts (The Experts Problem)

Suppose there are N experts (actions)

For $t = 1, 2, \dots, T$:

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss of every expert $\ell_t = (\ell_t(1), \dots, \ell_t(N)) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle = \mathbb{E}_{i \sim p_t} [\ell_t(i)]$

A special case of Online Convex Optimization with

$\mathcal{X} = \Delta_N$ and $f_t(p) = \langle p, \ell_t \rangle$

$$\text{Regret} := \sum_{t=1}^T \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t=1}^T \ell_t(i)$$

Follow the Leader (FTL)

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Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

Define
$$L_t(i) := \sum_{s=1}^t \ell_s(i)$$

“Follow the leader” strategy: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$ and let $p_t = e_{i_t}$

Follow the Leader (FTL)

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

$$\text{Regret} = \underbrace{\sum_{t=1}^T \langle p_t, \ell_t \rangle}_{\geq T-1} - \underbrace{\min_{i \in [N]} \sum_{t=1}^T \ell_t(i)}_{\approx \frac{T}{2}} \approx (T-1) - \frac{T}{2} \approx \Theta\left(\frac{T}{2}\right)$$

Softmin

Replacing “min” by “softmin”:

$$p_t(i) = \frac{w_t(i)}{\sum_{j=1}^N w_t(j)} \quad \text{where} \quad w_t(i) = e^{-\eta L_{t-1}(i)} \quad \textbf{(Exponential weights)}$$

“Follow-the-leader” (hardmin) is a special case when $\eta = \infty$

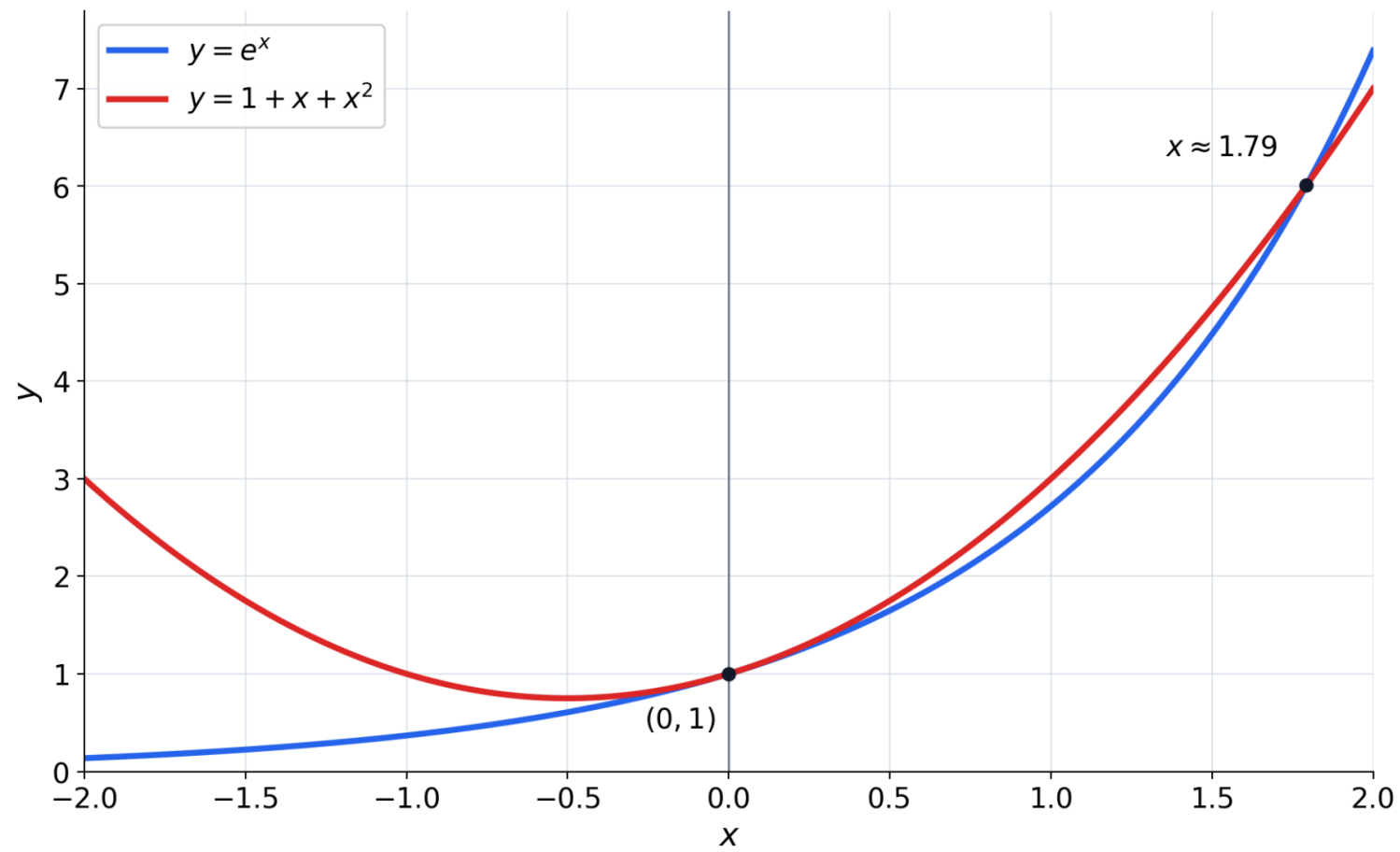
Regret Guarantee for Exponential Weights

Theorem 1.

Assume that $\eta \ell_t(i) \geq -1$ for all t, i . Then Exponential Weight ensures

$$\text{Regret} \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \sum_{i=1}^N p_t(i) \ell_t(i)^2$$

Proof of Theorem 1

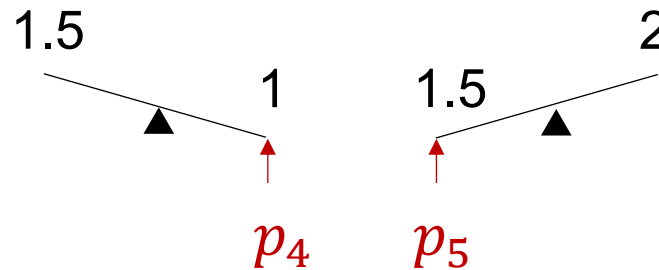
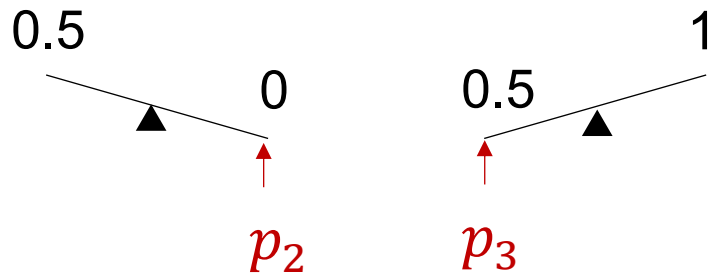


Why softmin works, while min does not?

The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

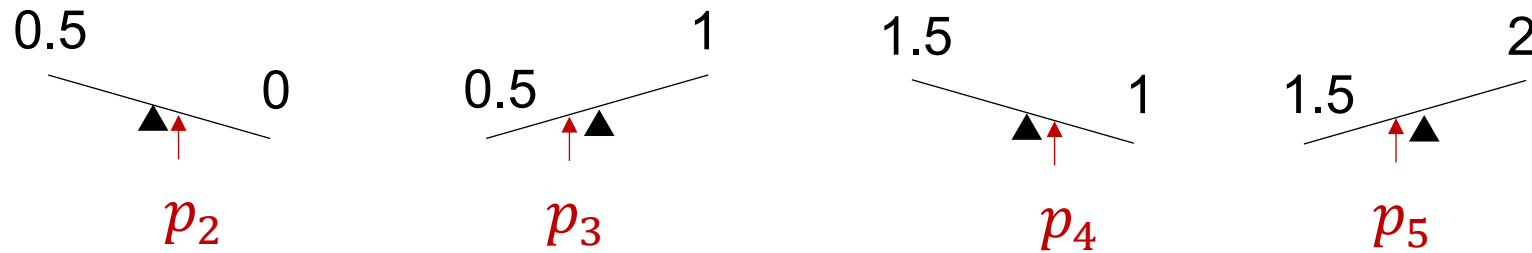
FTL: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$



The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

EW: $p_t(i) \propto e^{-\eta L_{t-1}(i)}$



Why Stability Helps?

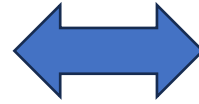
Follow The Regularized Leader

(Generalizing the Exponential Weights algorithm)

An Equivalent Form of EW

$$\psi(p) := \sum_{i=1}^N p(i) \ln p(i) \quad (\text{convex})$$

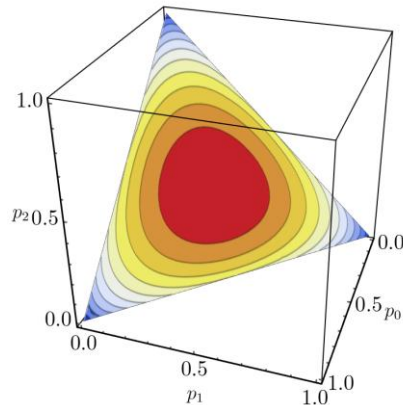
$$p_t(i) = \frac{e^{-\eta L_{t-1}(i)}}{\sum_{j=1}^N e^{-\eta L_{t-1}(j)}}$$



Verify yourself
(Lagrange multiplier)

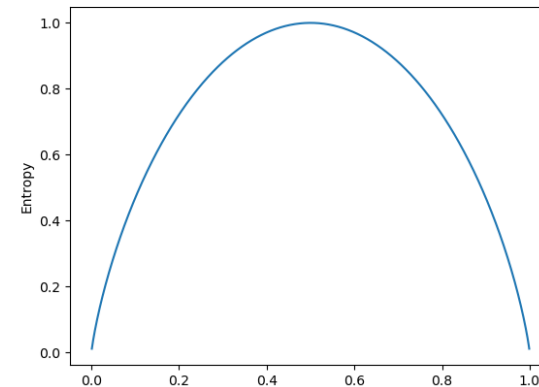
$$p_t = \operatorname{argmin}_{p \in \Delta_N} \left\{ \langle p, L_{t-1} \rangle + \frac{1}{\eta} \psi(p) \right\}$$

$-\psi(p) = \sum_{i=1}^N p(i) \ln \frac{1}{p(i)}$ is called the **(Shannon) entropy function** (concave)



N=3

[\(link\)](#)



N=2

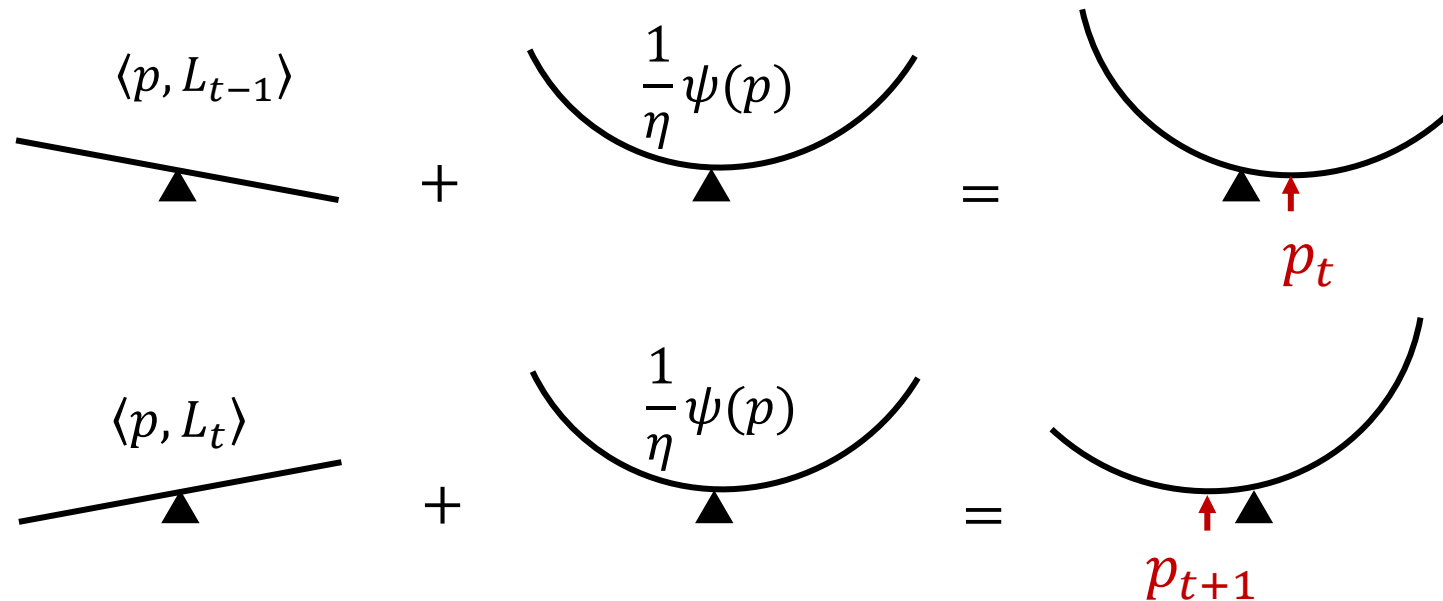
[\(link\)](#)

Follow the Regularized Leader (FTRL)

$$p_t = \operatorname{argmin}_{p \in \Delta_N} \left\{ \langle p, L_{t-1} \rangle + \frac{1}{\eta} \psi(p) \right\}$$

Regularizer

If ψ is *curved*, it makes the update more stable:



(Strongly) Convex Functions

Bregman Divergence

FTRL for Online Convex Optimization (OCO)

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Learner makes a decision $x_t \in \mathcal{X}$

Adversary decides a convex loss function $f_t: \mathcal{X} \rightarrow \mathbb{R}$

Adversary reveals f_t to the learner

Follow The Regularized Leader (FTRL)

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \sum_{s=1}^{t-1} f_s(x) + \frac{1}{\eta} \psi(x) \right\}$$

Regret Guarantee of FTRL in OLO

Theorem 2. Let ψ be μ -strongly-convex in norm $\|\cdot\|$.

Then FTRL in OCO ensures for any $x^* \in \mathcal{X}$,

$$\sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x^*) \leq \frac{\psi(x^*) - \min_{u \in \mathcal{X}} \psi(u)}{\eta} + \frac{\eta}{2\mu} \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2$$

$\|\cdot\|_*$ is the dual norm of $\|\cdot\|$

Special Case: Negative Shannon Entropy

$$\mathcal{X} = \Delta_N, \quad \psi(x) = \sum_{i=1}^N x(i) \log x(i)$$

Special Case: Euclidean Norm

$$\mathcal{X} = \text{general convex set}, \quad \psi(x) = \frac{1}{2} \|x\|_2^2$$

Proof of Theorem 2 (First-Order Optimality)

Proof of Theorem 2 (Potential Evolution)

OCO with only gradient feedback

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Learner makes a decision $x_t \in \mathcal{X}$

Adversary decides a convex loss function $f_t: \mathcal{X} \rightarrow \mathbb{R}$

Adversary reveals $\nabla f_t(x_t)$ to the learner

Online Mirror Descent

Online Mirror Descent

Implicit Online Mirror Descent

$$x_{t+1} = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ f_t(x) + \frac{1}{\eta} D_\psi(x, x_t) \right\}$$

Online Mirror Descent

$$x_{t+1} = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} D_\psi(x, x_t) \right\}$$

The regret guarantee and analysis are closely related to those of FTRL
(will be studied in Homework 1)

Special Case: Negative Shannon Entropy

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Special Case: Euclidean Norm

$$\mathcal{X} = \text{general convex set}, \quad \psi(x) = \frac{1}{2} \|x\|_2^2$$

Follow The Perturbed Leader

Online Linear Optimization with a Weaker Adversary

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Adversary decides a **linear** loss function $f_t(x) = \langle x, \ell_t \rangle$
(not revealing it to the learner at this moment)

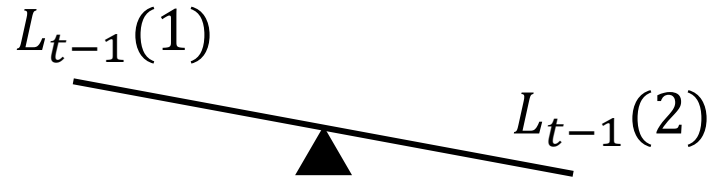
Learner makes a decision $x_t \in \mathcal{X}$

Adversary reveals ℓ_t to the learner



order swapped

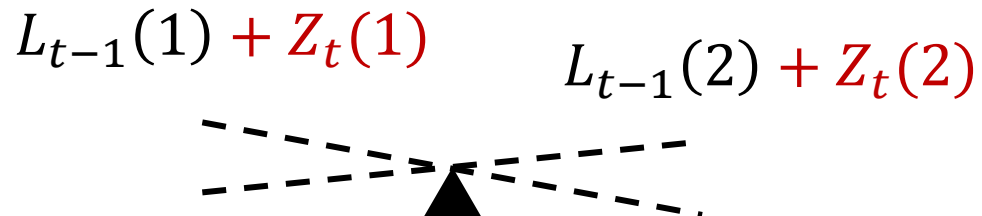
Another Way to Avoid Exploitation by Adversary



Follow the Leader: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$



As the learner will choose $i_t = 2$,
I'll make $\ell_t(2) \gg \ell_t(1) \dots$



$$Z_t(i) \sim \mathcal{N}(0, \sigma^2) \quad \forall i = 1, 2$$

Follow the Perturbed Leader:

$$i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i) + Z_t(i)$$



Hard to predict the learner's choice...

FTPL is also ensuring “stability”

FTPL for OLO

Follow The Perturbed Leader (FTPL) for OLO

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \sum_{s=1}^{t-1} \langle x, \ell_s \rangle + \langle x, z_t \rangle \right\}$$

where $z_t \in \mathbb{R}^d$ is drawn from some distribution $\mathcal{D}$.

$\mathcal{D}$ needs not be Gaussian, needs not be zero mean, and needs not be the same in every round.

When is FTPL more preferred than FTRL

For many common decision set $\mathcal{X}$, it is (computationally) much easier to solve

$$\operatorname{argmin}_{x \in \mathcal{X}} \langle x, L \rangle$$

then

$$\operatorname{argmin}_{x \in \mathcal{X}} \langle x, L \rangle + \psi(x)$$

for a given $L \in \mathbb{R}^d$.

When is FTPL more preferred than FTRL

Example: Online Shortest Path

Given: a graph, source, and sink.

The set of paths is denoted as $\mathcal{Q} \subset \{0,1\}^{|E|}$

For $t = 1, 2, \dots, T$:

Adversary decides the costs on all edges $\ell_t \in \mathbb{R}^{|E|}$

Learner chooses a path $q_t \in \mathcal{Q}$

Adversary reveals ℓ_t to the learner

$\mathcal{Q}$ above is not a convex set, but we can

- Run OCO in $\mathcal{X} = \text{convhull}(\mathcal{Q})$ (becoming a space of flows)
- Choose q_t such that $\mathbb{E}[q_t] = x_t$ in round t .

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